



Department of Information Technology

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CS3352 & Foundations of Data Science

Name of the Faculty member (s): Dr. V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / Regression towards the mean
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO7
- **Activity Chosen:** Jigsaw
- **Justification:**

The **Jigsaw cooperative learning strategy** is particularly suitable for teaching the statistical concept **Regression toward the mean**, as it transforms a complex and abstract idea into manageable, interconnected learning segments. The topic involves understanding statistical variability, correlation, prediction errors, and real-world applications, all of which can be effectively explored when students learn collaboratively.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The topic “**Regression Toward the Mean**” was taught in class using the Jigsaw activity method.
 - The class was divided into **seven home groups**, each consisting of nine students, formed based on academic performance and ability.
 - One student from each home group was selected as a **group leader**, and these leaders came together to form the **expert groups**, with one representative from each home group.
 - Each student in the expert group was assigned a specific subtopic related to Regression Toward the Mean, and **10 minutes** were provided for discussion and understanding.
 - The course instructor guided the students on how to carry out the activity effectively.
 - After the expert-group discussion, the groups were dissolved, and all students returned to their respective home groups.
 - In the home groups, every student acted as an **expert** in their assigned concept and shared it with their peers, ensuring that all members learned every part of the topic.
 - This collaborative approach enabled students to gain **in-depth understanding** of the concept. They later discussed and answered the questions provided, as illustrated in **Figure 1**.

Question:

1. Find the standard error of the estimate of the mean weight of high school football player

using the data given of weights of the players.

Player Number: 1 2 3 4 5 6 7 8 9 10

Weights in Pounds :150 203 176 190 168 193 189 178 197 1725

2. The values of x and their corresponding values of y are presented below:

x	0.5	1.5	2.5	3.5	4.5	5.5	6.5
y	2.5	3.5	5.5	4.5	6.5	8.5	10.5

(i) Find the least square regression line $y=ax+b$

(ii) Estimate the value of y when $x=10$.

3. Given the following dataset of employee, using regression analysis, find the expected salary of an employee if the age is 48.

Age	Salary
52	48500
46	46450
39	38750
41	41500
59	55500

4. Consider the following dataset with one response variable y and to predictor variables x_1 and x_2

y	140	155	159	179	192	200	212	215
X_1	60	62	67	70	71	72	75	78
X_2	22	25	24	20	15	14	14	11

Fit a multiple linear regression model to this dataset

5. Given the following dataset of employee, using regression analysis, find the expected salary of an employee if the age is 45.

Age	Salary
54	67000
42	43000
49	55000
57	.
35	25000

6. Calculate the correlation coefficient for the heights 'in inches' of fathers(x) and their son's(y) with the data presented below.

x	66	68	68	70	71	72	72
y	68	70	69	72	72	72	74

7. Estimate whether the following pairs of scores for X and Y reflect a positive relationship, a negative relationship, or no relationship. Hint: Note any tendency for pairs of X and Y scores to occupy similar or dissimilar relative locations.

X	Y
64	66
40	79
30	98
71	65
55	76

31	83
61	68
42	80
57	72

(b) Construct a scatterplot for X and Y. Verify that the scatterplot does not describe a pronounced curvilinear trend.

(c) Calculate r using the computation formula

• **Images / Screenshot of the practice:**



Figure 1: Students Iptika D and her team members presented their question and solution

- CO – PO / PSO mapping: PO / PSO mapped:

CO	PO1	PO2	PO3	PO4	PO12	PSO1	PSO2	PSO3
CO3	3	3	2	2	2	2	3	2

Innovative practice	PO1	PO2	PO3	PO4	PO12	PSO1	PSO2
	3	3	2	2	2	3	2
Justification for correlation	Apply basic Knowledge and fundamentals of Mathematics for solving practical problems	Identify the need of describing data for developing applications	Able to design solutions for applications to describe regression towards mean	Able to analyze the relationship between data and provide valid conclusions based on regression towards mean	Exhibit knowledge in regression towards mean to provide solution for complex computational problems	Exhibit knowledge in regression towards mean in the field of IoT, Cloud Computing and Data Analytics to provide IT solutions.	Exhibit knowledge in regression towards mean in software projects to deliver a quality product

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students reported that the Jigsaw activity made the complex concept of **Regression Toward the Mean** easier to understand, as the topic was divided into smaller, manageable parts.
- They appreciated the opportunity to become “experts” in a subtopic, which helped strengthen their confidence and deepen conceptual understanding.
- Many students felt that peer teaching was highly effective, as explanations given by classmates were more relatable and easier to follow.
- Students expressed that the activity improved their **communication skills, teamwork, and critical thinking abilities**.
- The activity demonstrated improved **student–student interaction**, enabling better clarification of doubts.

- ❖ ***Benefit of the practice:***

- Regression toward the Mean is a concept that involves correlation, variability, and prediction. The Jigsaw approach breaks the topic into smaller subtopics, making it easier for students to understand each component thoroughly.
- Instead of passively listening to a lecture, students actively engage in learning by becoming “experts” in a specific part of the topic. This active involvement deepens comprehension and retention.
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- Sharing ideas within expert and home groups helps students improve articulation, listening skills, and confidence in group discussions

- ❖ ***Challenges faced in implementation:***

- Regression toward the mean is mathematically and conceptually challenging. Some students initially struggled to grasp the idea even before joining their expert groups, requiring additional clarification from the instructor.
- The 10-minute discussion period in expert groups was not sufficient for some students to fully understand their assigned subtopics. Managing transitions between expert and home groups also consumed additional time.

References:

- ❖ <https://www.jigsaw.org/>

- ❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/VA_CS3352_Jigsaw.pdf

Signature of Faculty Member

(Dr.V. Anusuya)

HOD